

2. The eigenvalue-eigenvector problem

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Lecture due Feb 14, 2024 20:30 IST

In the course *Differential equations: 2 by 2 systems*, we learned that the first step in solving a linear 2×2 system of differential equations, $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$, is to find the eigenvalues and eigenvectors of the 2×2 matrix $\mathbf{A}$. The procedure for solving linear $n \times n$ systems of DEs is the same, and starts with finding eigenvalues and eigenvectors. And even outside the context of differential equations, the eigenvalues and eigenvectors of a matrix tell us a lot about what the matrix does as a function on $\mathbb{R}^n$.

Definition 2.1 Let $\mathbf{A}$ be an $n \times n$ matrix.

- An **eigenvalue** of $\mathbf{A}$ is a **scalar** λ such that $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ for some **nonzero** vector $\mathbf{v}$.
- An **eigenvector** of $\mathbf{A}$ **associated with an eigenvalue** λ is a **nonzero** vector $\mathbf{v}$ such that $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$.

(We also say that an eigenvector $\mathbf{v}$ “corresponds to,” or “belongs to” an eigenvalue λ .)

The eigenvalue-eigenvector problem is to find all possible scalars λ , and for each λ , all vectors $\mathbf{v}$ such that

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}.$$

Warning: Eigenvalues and eigenvectors are defined only for **square** matrices.

Warning: The zero vector $\mathbf{0}$ is never an eigenvector.

Note: Everyone allows that $\lambda = 0$ can be an eigenvalue.

Example 2.2 The matrix $\begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix}$ satisfies

$$\begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix} \mathbf{v} = 5\mathbf{v} \quad \text{for all } \mathbf{v} \text{ in } \mathbb{R}^3.$$

Therefore, the number **5** is an eigenvalue and all (nonzero) vectors in $\mathbb{R}^3$ are eigenvectors associated to the eigenvalue **5**.

Example 2.3 The diagonal matrix $\begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$ satisfies the following:

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = 0 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = -1 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

The first equation above shows that the scalar **2** is an eigenvalue with associated eigenvector $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$.

Similarly, the second and third equations show that **0** and **−1** are both eigenvalues, and $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ and

$\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ are eigenvectors associated to **0** and **−1** respectively.

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Example 2.4 The matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ satisfies

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$
$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} = (-1) \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Therefore, the scalar **1** is an eigenvalue with an associated eigenvector $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$, and **−1** is an eigenvalue with an associated eigenvector $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

Eigenvalue and eigenvector concept check

1/1 point (graded)

Let $\mathbf{A} = \begin{pmatrix} 1 & -1 & 2 \\ 0 & 3 & -4 \\ 2 & 1 & 0 \end{pmatrix}$. Given that $\mathbf{v} = \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix}$ is an eigenvector, find the eigenvalue λ that $\mathbf{v}$ is associated with.

$\lambda =$ ✓ Answer: 3

Solution:

The calculation

$$\mathbf{A}\mathbf{v} = \begin{pmatrix} 1 & -1 & 2 \\ 0 & 3 & -4 \\ 2 & 1 & 0 \end{pmatrix} \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} -3 \\ 6 \\ 0 \end{pmatrix} = 3\mathbf{v},$$

shows that $\mathbf{v}$ is an eigenvector associated with eigenvalue **3**.

You have used 2 of 3 attempts

i Answers are displayed within the problem

Scalar multiples of eigenvectors

1/1 point (graded)

As above, let $\mathbf{A} = \begin{pmatrix} 1 & -1 & 2 \\ 0 & 3 & -4 \\ 2 & 1 & 0 \end{pmatrix}$. We know that $\mathbf{v} = \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix}$ is an eigenvector associated to an eigenvalue λ (which you found in the previous problem).

Which of the following is true about the vector $\mathbf{w} = 2\mathbf{v}$?

- ☐ $\mathbf{w}$ is not an eigenvector.
- ☒ $\mathbf{w}$ is an eigenvector corresponding to the **same** eigenvalue λ .
- ☐ $\mathbf{w}$ is an eigenvector corresponding to an eigenvalue **different** from λ .



Solution:

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$$\mathbf{Aw} = \mathbf{A} (2\mathbf{v}) = 2 (\mathbf{Av}) = 2 (3\mathbf{v}) = 6\mathbf{v} = 3\mathbf{w},$$

so $\mathbf{w}$ is an eigenvector associated to the same eigenvalue, 3 .

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